

Title: Neuro-Geometric Recursive VLA : Topological invariant learning for grounded intelligence

Applicant: Akson Sam Varghese

Supervisor:

Keywords: Geometric Deep Learning, Embodied AI, Persistent Homology, Lie Group Transformations, SE(3) Equivariant World Models, Differentiable Topology Layers, Recursive Architecture, Topological Invariants

Abstract:

Current Vision-Language-Action (VLA) models struggle to generalise across dynamic environments, primarily due to the lack of physical grounding in hallucination-prone Large Language Models (LLMs). This proposal introduces the Neuro-Geometric Recursive VLA (NGR-VLA), a dual-system framework designed to ground agent behaviour in topological invariants — fundamental structural constraints such as connectivity and object permanence. System 1 functions as a topological world model, projecting inputs into a latent simplicial complex to learn the structural dynamics of the environment. System 2 acts as an invariant engine, refining action plans via iterative latent reasoning to ensure geometric consistency. By explicitly prioritising resource efficiency, this framework aims to establish a scalable, physically grounded foundation for embodied machine intelligence.

Introduction:

The convergence of large-scale language modelling with robotic control has established the VLA paradigm as the standard for embodied intelligence [1, 2]. However, VLA models relying on LLM backbones face critical limitations that hinder reliable deployment. These include the symbol grounding problem and the embodiment gap, along with severe inference latency, physical hallucinations, safety risks, data scarcity, and semantic fragility [1–7].

While recent works have attempted to mitigate these issues [3, 5, 8–11], current solutions remain resource-intensive and impractical for general use [3, 5, 10]. Implementations that address resource constraints often utilise higher-dimensional latent spaces to improve performance [12, 13], but they frequently suffer from degraded spatial reasoning in dynamic environments. Furthermore, mainstream geometry-first implementations rely on dense 3D/4D reconstruction, necessitating vast amounts of labelled data [8, 12, 15]. These metric-heavy approaches are computationally expensive and unstable; they struggle to adapt to dynamic environments where precise metric reconstruction is prone to failure.

Methodology:

I propose shifting the grounding mechanism from a metric geometry-first approach to topological invariance learning. I hypothesise that collaborative agents require a fundamental understanding of structure such as connectivity, containment, and solidity. The objective is to develop a framework that is both sample-efficient and computationally lightweight. Inspired by decoupled systems for efficient world modelling [1, 3, 14], I propose a dual-system framework comprising two core research objectives executed via System 1 (Perception) and System 2 (Action).

The first objective is to develop a perception architecture capable of projecting inputs, such as ego-centric videos, hierarchical DOM trees for web browsers, into a latent simplicial complex space. This is achieved by System 1, which utilises a vision encoder (e.g., ViT, MambaVision) to project data into a latent space

structured by persistent homology and regulated via a topological loss [16]. Inspired by the regularised methods of V-JEPA [14], System 1 could employ topologically-regularised masked modelling in a self-supervised regime. This enables the agent to learn physical laws (e.g., object permanence) from raw, unlabelled interaction sequences by predicting the connectivity of occluded regions.

The second objective is to enable efficient, real-time decision-making using parameter-efficient iterative architectures (e.g., Tiny Recursive Model, Mamba SSM) [17, 19]. I aim to investigate whether iterative latent reasoning can approximate complex action dynamics with a smaller number of parameters and lower memory footprint as compared to current systems. System 2 utilises this to recursively refine the agent's action plans, verifying their geometric or structural consistency via manifold constraints (e.g., Lie Group Transformations (LGT) for robotics, Graph Validity for web agents) [18, 25]. This ensures that generated actions respect the kinematic constraints of the embodiment (rigid-body dynamics) before execution.

Evaluation:

To rigorously and accurately validate the NGR-VLA, I propose an evaluation framework that measures mathematical stability, logical safety, and functional utility.

The first metric is an implementation of a topological consistency score intended to measure mathematical stability, to quantify the invariance of the latent space to non-structural noise (e.g., camera rotation, lighting changes, visual appearances in browsers). This could be achieved by computing the Bottleneck or Wasserstein distance between the persistence diagrams of a canonical object and its deformed variants. The expectation is a low variance in distance metrics, indicating that the model has learned robust topological features that survive visual perturbations [20-22].

The second metric is to perform logical consistency checks intended to measure symbolic safety and to ensure the agent's understanding of the world adheres to fundamental physical laws of the environment. This requires development of a symbolic auditor based on logical axioms (e.g., An object cannot be in two disjoint locations simultaneously). The latent graph is discretised into predicates and checked against these axioms [23].

The third metric is to perform layout invariance tests, intended to validate generalisation across domain shifts. The method evaluates the agent on standardised benchmarks for digital and physical agents (e.g., VisualWebArena for changing CSS layouts, ManiSkill2 for deformable objects), validating that the agent relies on the invariant task structure rather than brittle visual correlations [24].

Execution Plan:

The proposed execution plan is outlined below:

In Year 1, I will develop the simplicial projection module (System 1) and validate mathematical stability on datasets (e.g., synthetic 3D datasets like ShapeNet). In Year 2, I will develop the recursive model (System 2) and symbolic auditor to validate layout invariance on digital domain datasets (e.g., WebArena). In Year 3, I will experiment on different environments: robot simulation, video games and web environments, to assess and improve the robustness of the system. In Year 4, I will conduct final benchmarking against SOTA baselines and conclude the findings in the thesis and present findings via conferences.

References:

- [1] Fung, P., Bachrach, Y., Celikyilmaz, A., Chaudhuri, K., Chen, D., Chung, W., ... & Malik, J. (2025). Embodied AI agents: Modeling the world. arXiv preprint arXiv:2506.22355.
- [2] Zitkovich, B., Yu, T., Xu, S., Xu, P., Xiao, T., Xia, F., ... & Han, K. (2023, December). Rt-2: Vision-language-action models transfer web knowledge to robotic control. In Conference on Robot Learning (pp. 2165-2183). PMLR.
- [3] Bjorck, J., Castañeda, F., Cherniadev, N., Da, X., Ding, R., Fan, L., ... & Zhu, Y. (2025). Gr00t n1: An open foundation model for generalist humanoid robots. arXiv preprint arXiv:2503.14734.
- [4] Zhang, D., Sun, J., Hu, C., Wu, X., Yuan, Z., Zhou, R., ... & Zhou, Q. (2025). Pure vision language action (vla) models: A comprehensive survey. arXiv preprint arXiv:2509.19012.
- [5] Huang, X., Chen, T., Bryson, M., Chaffey, T., Chen, H., Choo, K. K. R., & Manchester, I. R. (2025). Trust in LLM-controlled Robotics: a Survey of Security Threats, Defenses and Challenges. arXiv preprint arXiv:2601.02377.
- [6] Zhang, B., Li, J., Shen, J., Cai, Y., Zhang, Y., Chen, Y., ... & Yang, Y. (2025). VLA-Arena: An Open-Source Framework for Benchmarking Vision-Language-Action Models. arXiv preprint arXiv:2512.22539.
- [7] Perlo, J., Robey, A., Barez, F., Floridi, L., & Mökander, J. (2025). Embodied ai: Emerging risks and opportunities for policy action. arXiv preprint arXiv:2509.00117.
- [8] Li, X., Heng, L., Liu, J., Shen, Y., Gu, C., Liu, Z., ... & Dong, H. (2025). 3ds-vla: A 3d spatial-aware vision language action model for robust multi-task manipulation. In the 9th Annual Conference on Robot Learning.
- [9] Bhat, V., Lan, Y. H., Krishnamurthy, P., Karri, R., & Khorrami, F. (2025). 3D CAVLA: Leveraging Depth and 3D Context to Generalize Vision Language Action Models for Unseen Tasks. arXiv preprint arXiv:2505.05800.
- [10] Liu, H., Long, J., Wu, J., Hou, J., Tang, H., Jiang, T., ... & Yao, W. (2025). Eva-VLA: Evaluating Vision-Language-Action Models' Robustness Under Real-World Physical Variations. arXiv preprint arXiv:2509.18953.
- [11] Shao, R., Li, W., Zhang, L., Zhang, R., Liu, Z., Chen, R., & Nie, L. (2025). Large vlm-based vision-language-action models for robotic manipulation: A survey. arXiv preprint arXiv:2508.13073.
- [12] Ni, C., Chen, C., Wang, X., Zhu, Z., Zheng, W., Wang, B., ... & Mei, W. (2025). SwiftVLA: Unlocking Spatiotemporal Dynamics for Lightweight VLA Models at Minimal Overhead. arXiv preprint arXiv:2512.00903.
- [13] Patratskiy, M. A., Kovalev, A. K., & Panov, A. I. (2025). Spatial traces: Enhancing vla models with spatial-temporal understanding. Optical Memory and Neural Networks, 34(Suppl 1), S72-S82.
- [14] Assran, M., Bardes, A., Fan, D., Garrido, Q., Howes, R., Muckley, M., ... & Ballas, N. (2025). V-jepa 2: Self-supervised video models enable understanding, prediction and planning. arXiv preprint arXiv:2506.09985.
- [15] Hu, W., Lin, J., Long, Y., Ran, Y., Jiang, L., Wang, Y., ... & Pang, J. (2025). G2 VLM: Geometry Grounded Vision Language Model with Unified 3D Reconstruction and Spatial Reasoning. arXiv preprint arXiv:2511.21688.
- [16] Zia, A., Khamis, A., Nichols, J., Tayab, U. B., Hayder, Z., Rolland, V., ... & Petersson, L. (2024). Topological deep learning: a review of an emerging paradigm. Artificial Intelligence Review, 57(4), 77.
- [17] Jolicoeur-Martineau, A. (2025). Less is more: Recursive reasoning with tiny networks. arXiv preprint arXiv:2510.04871.

- [18] Simeonov, A., Du, Y., Tagliasacchi, A., Tenenbaum, J. B., Rodriguez, A., Agrawal, P., & Sitzmann, V. (2022, May). Neural descriptor fields: SE(3)-equivariant object representations for manipulation. In 2022 International Conference on Robotics and Automation (ICRA) (pp. 6394-6400). IEEE.
- [19] Gu, A., & Dao, T. (2024, May). Mamba: Linear-time sequence modeling with selective state spaces. In First conference on language modeling.
- [20] Alliez, P., Cohen-Steiner, D., Tong, Y., & Desbrun, M. (2007, July). Voronoi-based variational reconstruction of unoriented point sets. In Symposium on Geometry processing (Vol. 7, pp. 39-48).
- [21] Carrière, M., Chazal, F., Ike, Y., Lacombe, T., Royer, M., & Umeda, Y. (2020, June). Perslay: A neural network layer for persistence diagrams and new graph topological signatures. In the International Conference on Artificial Intelligence and Statistics (pp. 2786-2796). PMLR.
- [22] Atienza, N., González-Díaz, R., & Soriano-Trigueros, M. (2018). On the stability of persistent entropy and new summary functions for TDA. arXiv preprint arXiv:1803.08304.
- [23] Liang, B., Wang, Y., & Tong, C. (2025). AI Reasoning in the Deep Learning Era: From Symbolic AI to Neural-Symbolic AI. *Mathematics*, 13(11), 1707.
- [24] Diamantis, I. (2025). The shape of data: Topology meets analytics. A practical introduction to topological analytics and the Stability Index (TSI) in business (arXiv:2511.13503).
- [25] Wu, J., Zhao, Q., Chen, Z., Qin, K., Zhao, Y., Wang, X., & Yao, Y. (2025). GAP: Graph-Based Agent Planning with Parallel Tool Use and Reinforcement Learning. arXiv preprint arXiv:2510.25320.